Normalization

- It helps reduce redundancy. → we take bigger table and break it into smaller tables.

Functional Dependencies

[Relationship b/w 2 attributes
One of them is mostly primary / unique key & other one is mostly non-prime or non-key attribute

$X \longrightarrow Y$

Primary Key (Determinant) — Non-Primary Key (Dependent)

Employee Table		
Emp-ID	Emp-Name	Emp-Address

① Trivial functional Dependency

$A \rightarrow B$

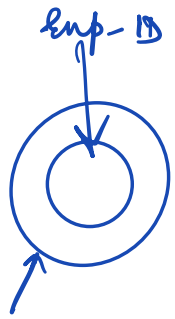
[If B is subset of A]

$A \rightarrow A$
$B \rightarrow B$

Trivial Dependencies

eg {Emp-ID, Emp-Name} → Emp-ID

{ " " } → Emp_Address



Emp-Address is not trivially functionally Dependent on {Emp-ID, Emp-Name} as it is not subset of this superset.

Non-Trivial

$A \rightarrow B$

B should not be subset of A.

Emp-ID → Emp Name

Augmentation Rule states that addition of the same attributes to the right and left side will result in another valid dependency.

→ If B can be determined from A, then adding an attribute to this FD won't change anything.

Why Nor..?
ANOMALIES → Insert, Update, Delete

Employee Table

Emp. ID	Emp. Name	Emp. Address	Attendance
1	Ankush	~~Patna~~ Delhi	P
1	Ankush	Patna	P

↳ Keeps happening → updation anomaly → If we want to update address of Ankush to Delhi - now we've have to update every address.

Employee Table

Emp-ID	Emp-Name	Emp-Address	DOJ	DOT (not needed)
1	Ankush	Delhi	1/1/20	NULL
				NULL

Since we'll need it for almost every employee → So this is insertion anomaly

Employee Table

Emp-ID	Emp-Name	Emp-Address	Dept
~~1~~	~~Ankush~~	~~Delhi~~	~~Sales~~
2	Prakash	Delhi	Marketing
3	Suvina	Delhi	Marketing

→ Now this important Sales Dept is also deleted from database → This is Deletion anomaly.

[We just wanted to delete Ankush's data]

1NF

Normal forms are actually used to reduce or eliminate redundancy in the data.

- 1st Normal form
- 2nd Normal form
- 3rd Normal form
- Boyce Codd form (BCNF) → 3.5NF

1st Normal form

- If a relation contains multivalued attribute, it violates 1NF.

Emp_details

Emp-ID	Emp-Name	Emp-Phone	Emp-Dept
1	Ankush	888888	Sales
2	Bina	89888 898 on	Marketing

Since its multivalued - it violates 1NF.

Options →

① → We can divide the table into two

Emp-ID	Emp-Phone

Emp-ID	Emp-Name	Emp-Dept

② We record it as 2 tuples → follows 1st NF.

Emp-details

Emp-ID	Emp-Name	Emp-Phone	Emp-Dept
1	Ankush	888888	Sales
2	Bina	89888	Marketing
2	Bina	898 ~	Marketing

But we can't set Emp-ID as primary key.

To satisfy 1st NF → 4 conditions

1st NF
↓
2nd NF
↓
3rd NF
↓
BCF

① Each attribute should contain atomic values.

② Each value stored in an attribute should be of same type

③ All attributes in a table should've unique names

④ The order of the data stored in the table doesn't matter.

2NF

Conditions → ⓐ Relation must be in 1NF

ⓑ There should not be any Partial Dependency.

dependent

Student Table

Stu-id	St-name	Phone	Address
1	Anmol	8888	Delhi
2	Bina	9990	Delhi
3	Carl	1000	Patna

Partial Dependency →

Composite Key → Primary Key (71 attributes)

Course Relation

CS 101 → Mr A
CS 102 → Mr B
CS 103 → Mrs C

Cou-id	Stu-id	Marks	~~Prof-Name~~
CS101	1	10	Mr. A
CS102	1	9	Mr. B
CS103	1	8	Mrs C
CS101	2	9	Mr A

how many marks stud-id = 1 got? → can't tell

" " " student got in CS102? → " "

which student

Thus, Primary Key → Cou-ID + Stu-ID
(to identify marks)

Q → Which course does Mr. A take?

We don't need Stu-ID anymore → just Cou-ID

Thus, its partial dependency

If one part of composite key can independently determine another attribute — this is called Partial Dependency.

What to do?

We delete Prof-Name from course relation

Prof Table

Cou-ID	Prof-Name
=	=

Conditions:

3NF

→ Relation must be in 2NF

→ There should not be any Transitive Dependency

If $a = b$ and $b = c$ } Transitive Property of Mathematics

Then $\boxed{a = c}$

If $A \rightarrow B$ and $B \rightarrow C$ then $A \rightarrow C$

Determinant

Dependent

Transitive Dependency on functions

Student Table

ID	Name	State	~~Country~~
1	Atul	Bihar	~~India~~
2	Bina	Texas	~~USA~~

Create new table Address

State	Country
=	2

A B

ID → State

B C

State → Country

A → C

→ Yes we can

derive ID → Country

as ID is primary key

Shortcut to determine : If an attribute is dependent on non-prime attribute, then there might be transitive dependency.

Normalization – BCNF (Boyce-Codd Normal form) / 3.5 Normal form

Conditions → 3NF

→ If A → B, A should be super key

A ↓ Non-Primary Key

B ↓ B cannot be primary key

Course

Stu-ID	Course-ID	Prof. Name
1	CS101	Mr A
2	CS101	Mrs C

Primary Key : Stu-ID + Course-ID

Professor → Cou-ID

Professor ↓ Non-Primary

Cou-ID ↓ Part of Primary Key.

Not in BCNF

Solution

Stu-ID	Prof-ID

Prof-ID	Course-ID	Prof-Name